Roll — Frequency sweep (chirp)
Recorded straight-flight SITL; normal controllers remain active

Normalized excitation
DDS requested
Applied logical excitation

Commanded PWM (µs)
SERVO1 baseline
SERVO1 final

Angular rate (rad/s)
p: roll
q: pitch
r: yaw

Attitude change (deg) from maneuver start
Roll
Pitch
Yaw

Airspeed (m/s)
Airspeed

Time from first enabled command (s); shaded = commanded maneuver